Chenmu Zhang

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SUMMARY

Computational materials scientist (Ph.D., UT Austin; postdoc, Rice) building autonomous LLM agent systems for computational materials science.

Developed MatClaw, an LLM agent built on a persistent Python kernel and designed for data analysis and long-horizon workflows on HPC clusters.

Applied LLM agents in real research: a materials-discovery study run end to end by MatClaw (manuscript in preparation), and an auto-research loop that produced the top from-scratch band-gap model on the public MatBench leaderboard (arXiv 2606.29717, COLM 2026 workshop paper).

Built CompMat-Bench, a benchmark of 94 tasks drawn from recently published computational materials studies, quantifying AI-agent capability in real research work (arXiv 2610.00636, under review).

RESEARCH EXPERIENCE

► **Postdoctoral Research Associate** May 2023 – Present

Advisor: Prof. Boris I. Yakobson, Rice University, Houston, TX

- Designed and built MatClaw, an open-source code-first LLM agent that carries out long-running simulation workflows on HPC clusters (arXiv 2604.02688).
- Applied MatClaw to a materials-discovery study of off-stoichiometric Cu–In–Se, where the agent ran generative structure prediction, interatomic-potential screening and DFT validation end to end (co-first-author manuscript, in preparation).
- Applied autonomous research to neural-network design for band-gap prediction: an LLM agent redesigned, trained and evaluated the model over 184 experiments with no human input; the final model ranks first among from-scratch models on the public MatBench leaderboard, ahead of all 17 expert-designed ones (arXiv 2606.29717, COLM 2026 workshop paper).
- Designed CompMat-Bench, a benchmark of tasks derived from recently published studies with rule-based grading against reproduced results; evaluated four agents on three LLMs under single-task and workflow conditions (arXiv 2610.00636, under review).

► **Graduate Research Assistant** Aug. 2018 – Apr. 2023

Advisor: Prof. Yuanyue Liu, University of Texas at Austin, Austin, TX

- First-principles electron-phonon transport in 2D semiconductors: established why 2D semiconductors generally have low mobility and which electronic structures escape it, guiding the discovery of room-temperature high-mobility 2D semiconductors (PRL 2020, 2023); extended the theory to substrate, surface and remote-phonon scattering (ACS Nano 2024; Nano Lett. 2026).

PROJECTS

- **MatClaw** — LLM agent for materials computations, built for data analysis and long-horizon work: persistent Python kernels keep models, trajectories and running jobs in memory across steps; it writes its own tools, and any code dropped into its workspace becomes one; it runs and supervises hours-long simulations on remote HPC clusters. Open source; sole developer.
- **AutoHive** (in development) — multi-agent framework for autonomous research on materials: several LLM agents work in parallel, each editing, training and evaluating models that predict materials properties and keeping what improves them; agents can share their records with one another. Current work: how information sharing between agents changes research efficiency. Sole developer.

► **CompMat-Bench** (code release upon publication) — benchmark that measures how well AI agents perform real computational materials research: tasks come from recently published studies, and the expensive simulations are reproduced in advance, so evaluating an agent is cheap and repeatable, with grading by fixed rules rather than an LLM judge.

SKILLS

LLM agents: multi-agent systems; tool-use agents; code generation and execution by agents; long-horizon task execution; closed-loop computational discovery; retrieval-augmented generation; HPC job orchestration; agent integration with simulation codes.

Agent evaluation: agent benchmark design and evaluation on materials research tasks; rule-based grading; pass-rate, cost and reliability metrics.

Materials simulation: DFT (VASP, Quantum ESPRESSO), molecular dynamics (LAMMPS), electron-phonon transport and phonons (Wannier90, Phonopy), materials informatics and high-throughput workflows (pymatgen, ASE, atomate2, jobflow), machine-learned interatomic potentials and generative models (MACE, MatterSim, MatterGen).

SELECTED PUBLICATIONS

18 publications (13 first-author), including *Phys. Rev. Lett.*, *J. Am. Chem. Soc.*, *ACS Nano*, and *Nano Lett.*; 613 citations, h-index 10 (Google Scholar, Aug. 2026). Full list: Google Scholar.

1. **Zhang, C.**; Yakobson, B. I. MatClaw: An Autonomous Code-First LLM Agent for End-to-End Materials Exploration. arXiv preprint arXiv:2604.02688, **2026**.
2. **Zhang, C.**; Felix, L. C.; Zhang, J.-J.; Li, X.; Jiang, X.; Jiang, T.; Mishra, S.; Qin, X.; Yakobson, B. I. CompMat-Bench: Benchmarking AI Agents for Computational Materials Science. arXiv preprint arXiv:2610.00636, (Submitted) **2026**.
3. **Zhang, C.**; Yakobson, B. I. Optimizing Expert-Designed Crystal Graph Networks for Band-Gap Prediction with an Autonomous LLM Research Loop. arXiv preprint arXiv:2606.29717, accepted at the Workshop on Language Models for Scientific Discovery (LM4Sci), COLM **2026**.
4. Yuan*, Z.; **Zhang*, C.** Off-Stoichiometric Energetic and Electronic Asymmetry in Ternary Cu–In–Se: Task-Oriented Materials Discovery by an Autonomous LLM Agent. (In preparation; * equal contribution) **2026**.
5. **Zhang, C.**; Liu, Y. Enhancing Carrier Mobility by Remote Phonons. *Nano Letters* **2026**, 26(30), 9716–9721.
6. **Zhang, C.**; Liu, Y. Electron-Surface Scattering from First-Principles. *ACS Nano* **2024**, 18(40), 27433–27439.
7. **Zhang, C.**; Wang, R.; Mishra, H.; Liu, Y. Two-Dimensional Semiconductors with High Intrinsic Carrier Mobility at Room Temperature. *Physical Review Letters* **2023**, 130(8), 087001.
8. Cheng, L.; **Zhang, C.**; Liu, Y. Why Two-Dimensional Semiconductors Generally Have Low Electron Mobility. *Physical Review Letters* **2020**, 125(17), 177701.

EDUCATION

► **University of Texas at Austin, Austin, TX** Aug. 2018 – Apr. 2023

Ph.D. in Materials Engineering
Advisor: Prof. Yuanyue Liu
Thesis: *"Phonon-limited Carrier Transport in 2D Materials from First Principles"*

► **University of Science and Technology of China (USTC), Hefei, China** Sep. 2014 – Jul. 2018

B.S. in Physics
Advisor: Prof. Wenguang Zhu
Thesis: *"First Principles Calculations on 2D Ferroelectric Materials under Uniaxial Strain"*